MILITARY SPECIFICATION

CARTRIDGE ASSEMBLY & INSTRUCTOR'S CONTROL PANEL ASSEMBLY NIKE HERCULES TRAINER M-74

1. SCOPE

1.1 This specification covers requirements, quality assurance provisions, and the preparation for delivery criteria for one type of assembly entitled Cartridge Assembly & Instructor's Control Panel Assembly which is used with the Nike Hercules Trainer M-74.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-1168	- Lot Numbering of Ammunition.
MIL-STD-1169	- Packaging, Packing and Marking for Shipment of Inert Ammunition Components.
MIL-STD-1460	- Soldering of Electrical Connections and Printed Wiring Assemblies, Procedure For.

DRAWINGS

ORDNANCE CORPS

8846032	- Instructor's Control Panel Assembly.
8846038	- Cartridge Assembly.
8846157	- Instructor's Control Panel.
8846199	- Cartridge Support Assembly.
8846220	- Cartridge Shell Assembly.
EL-8855810	- Equipment List.

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(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials. The contractor shall furnish objective evidence (see 6.2) that the material complies with the specifications listed on the applicable drawing.

3.2 Assembly. The Cartridge Assembly & Instructor's Control Panel Assembly shall comply with all requirements specified on the drawing (dwg) 8846038 and 8846032, with all requirements specified herein, and with the requirements of all applicable specifications to the extent specified on the drawing and herein.

3.3 First article testing. Requirements for the submission of first article samples by the contractor shall be as specified in the contract (see 4.2).

3.4 Continuity. The Cartridge Assembly & Instructor's Control Panel Assembly shall exhibit point to point continuity and malfunction when tested as specified in 4.4.1.

3.5 Pneumatic. The Cartridge Assembly & Instructor's Control Panel Assembly shall respond to the pneumatic functions when tested as specified in 4.4.2.

3.6 Soldering. All soldering shall comply with the requirements specified in MIL-STD-1460.

3.7 Workmanship.

3.7.1 Plastic and metal characteristics. All components shall be free from cracks, splits, cold flow, shrinkage, cold shuts, inclusions, porosity, or any similar characteristic.

3.7.2 Threads. Threads shall be full and undamaged for the entire minimum length or depth as required on the applicable drawing.

3.7.3 Burrs. Parts shall be free from burrs.

3.7.4 Foreign matter. Parts shall be free from chips, dirt, grease, rust, corrosion, or any embedded foreign material.

3.7.5 Cleaning. The cleaning method shall not be injurious to any of the parts, nor shall the parts be contaminated by the cleaning agent.

3.7.6 Electrical. Cable assemblies and wires shall be dressed in such a manner as to prevent pinching or chafing of the wire insulation. Ends of wires shall be free from burrs, sharp edges, or points. All parts shall be free from flux, chips, dirt, grease, rust or foreign material.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 First article inspection.

4.2.1 Quantity to be submitted. When the contract requires first article inspection, the contractor shall submit 3 complete assemblies (see dwg 8846038 and 8846032) for first article testing to the testing facility designated in the invitation for bids or request for proposal.

4.2.2 Inspection and testing to be performed. The first article inspection sample shall be subjected to the inspections and tests specified in Table I. Tests shall be conducted in order specified in Table I.

Table I

First article inspection and tests

Inspection or Test	Dest	Non-Dest
1. Classification of Characteristics (see 4.3.2.1)		X
2. Continuity (see 3.4)		X
3. Pneumatic (see 3.5)		X

4.2.3 Inspection equipment. The inspection and testing specified in 4.2.2 shall be accomplished by using the gages and test equipment specified in 4.3.4.1.

4.2.4 Rejection. If any item fails to comply with any of the applicable requirements, the first article sample shall be rejected.

4.3 Quality conformance inspection.

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4.3.1 Lot formation.

4.3.1.1 Production lot. A lot shall consist of parts and assemblies produced by one manufacturer in one unchanged process, in accordance with the same drawing, same drawing revision, same specification and same specification revision. Drawing, specification and process changes not affecting safety, performance, storage, or interchangeability, as determined by the Government, shall not necessitate changing the lot interfix number. The criteria and procedures for the assignment of lot numbers shall be in accordance with MIL-STD-1168.

4.3.1.2 Inspection lot. A lot or batch and its formation, size, and presentation is described in MIL-STD-105, Section 5. Accordingly, a lot shall mean an inspection lot and a batch shall mean an inspection batch for the purpose of this specification. The manner in which each inspection lot or batch is to be presented and identified by the contractor, shall be designated or approved by the Government representative.

4.3.2 Product inspection examination. A sample shall be selected at random from each inspection lot in accordance with the applicable major or minor characteristic table (see Table II and Table III), and inspected for all characteristics in the applicable classification. Any characteristics in the sample found to be defective shall reject the inspection lot. Sequence for the inspection of characteristics in each classification need not be followed as long as all characteristic inspections are performed.

Table II
Major Characteristics

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-16	all	28-29	23	52-56	31	132-156	39
17	16	30-31	24	57-62	32	157-191	40
18	17	32-34	25	63-69	33	192-244	41
19-20	18	35-36	26	70-76	34	245-334	42
21	19	37-40	27	77-86	35	335-519	43
22-23	20	41-43	28	87-97	36	520-1120	44
24-25	21	44-47	29	98-112	37	over 1120	45
26-27	22	48-51	30	113-131	38		

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Table III
Minor Characteristics

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-9	all	14-16	12	28-33	16	77-121	20
10	9	17-18	13	34-41	17	122-273	21
11	10	19-22	14	42-54	18	over 273	22
12-13	11	23-27	15	55-76	19		

4.3.2.1 Classification of characteristics (see 3.2). Critical, major and minor characteristics are defined in MIL-STD-105. Inspection shall be conducted as follows:

4.3.2.1.1 Cartridge Support Assembly (see dwg 8846199) Prior to Assembly with Cartridge Shell Assembly.

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
101.	Servoed Synchro cover missing or not securely fastened to Cartridge Support	Visual
102.	Servoed Synchro Assemblies missing or not securely fastened to Cartridge Support	Visual
103.	Solenoid valve assembly missing	Visual
104.	Any operation missing	Visual
105.	Fuses missing or incorrect fuses in socket	Visual
106.	Evidence of poor workmanship (see 3.7)	Visual
107.	Insulation sleeving loose, torn or damaged	Visual
108.	Improper soldering (see 3.6)	Visual
109.	Servoed Synchro plugs not properly connected to cartridge harness receptacles	Visual
*110.	Height of pin(s) (pins of the same diameter must all appear to be within 1/16 inch in height relative to adjacent pins)	Visual/Gage
**111.	Depth of socket(s) (female sockets of the same diameter must all appear to be within 1/16 inch in height relative to adjacent pins)	Visual/Gage

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Major:

112.	Connector loose in Cartridge Support or connector cracked	Visual
113.	Insulation missing from harness wire	Visual
114.	Any wire broken	Visual
115.	Foreign matter in cartridge support unless removable	Visual
116.	Foreign matter on any pin unless removable	Visual
117.	Marking missing, misleading, or unidentifiable	Visual
118.	Painting defective to the extent it shows primer coat or bare metal	Visual
119.	Any pin bent or broken	Visual
120.	Relay missing or loose	Visual
121.	Capacitors missing or loose	Visual
122.	Connector deformed to extent it prevents assembly	Visual
123.	Objective evidence relating to material missing or improper (see 3.1)	Visual

Minor: None defined.

4.3.2.1.2 Cartridge Shell Assembly (see dwg 8846220) Prior to Assembly with Cartridge Support Assembly.

Categories	Characteristics	Method of Inspection

Critical: None defined.

Major:

101.	Flexible tube missing, cracked or split	Visual
102.	Pressurizing Adapter Assembly missing or loose	Visual
103.	Tube of Housing Adapter missing, cracked or split	Visual
104.	Foreign matter in Cartridge Shell unless removable	Visual
105.	Markings missing, misleading, or unidentifiable	Visual
106.	Painting defective to the extent it shows primer coat or bare metal	Visual
107.	Mounting holes on Cartridge Shell Ring improperly located or tapped	Gage

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Major:

108.	Mounting holes on Cartridge Shell Plate improperly located or drilled	Gage
109.	Plug Alignment Bracket missing	Visual
110.	Evidence of poor workmanship (see 3.7)	Visual
111.	Objective evidence relating to material missing or improper (see 3.1)	Visual

Minor: None defined.

4.3.2.1.3 Cartridge Assembly (see dwg 8846038).

Categories	Characteristics	Method of Inspection

Critical: None defined.

Major:

101.	Dust Covers missing from connectors on cartridge	Visual
102.	Fuseholder guard missing	Visual
103.	Evidence of poor workmanship (see 3.7)	Visual
104.	Markings missing, misleading or unidentifiable	Visual
105.	Painting defective to the extent it shows primer coat or bare metal	Visual
106.	Assembly punctured or cracked	Visual
107.	Objective evidence relating to material missing or improper (see 3.1)	Visual

Minor: None defined.

4.3.2.1.4 Instructor's Control Panel (see dwg 8846157) Prior to Assembly.

Categories	Characteristics	Method of Inspection

Critical: None defined.

Major:

101.	Mounting holes improperly located	Gage
102.	Mounting holes improperly drilled	Gage
103.	Mounting slots improperly located	Gage
104.	Mounting slots improperly cut	Gage

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Major:

105.	Panel deformed in any area	Visual
106.	Marking missing, misleading or unidentifiable	Visual
107.	Painting defective to extent it shows primer coat or bare metal	Visual
108.	Evidence of poor workmanship (see 3.7)	Visual
109.	Objective evidence relating to material missing or improper (see 3.1)	Visual

Minor: None defined.

4.3.2.1.5 Instructor's Control Panel Assembly (see dwg 8846032).

Categories	Characteristics	Method of Inspection

Critical: None defined.

Major:

101.	Any component missing or damaged	Visual
102.	Improper soldering (see 3.6)	Visual
103.	Any wire broken	Visual
104.	Insulation sleeving missing, loose, or torn	Visual
105.	Any operation missing	Visual
106.	Deformation in any area	Visual
107.	Evidence of poor workmanship (see 3.7)	Visual
*108.	Height of pin(s) (pins of the same diameter must all appear to be within 1/16 inch in height relative to the adjacent pins)	Visual/Gage
109.	Insulation missing from harness wire	Visual
110.	Foreign matter in Instructor's Panel Assembly unless removable	Visual
111.	Foreign matter or any pin unless removable	Visual
112.	Any pin missing, loose, bent or broken	Visual
113.	Tubing cracked or split	Visual
114.	Gage glass broken or cracked	Visual
115.	Tubing missing	Visual

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Major:
116. Objective evidence relating to material missing or improper (see 3.1).....Visual

Minor: None defined.

*In the event pin(s) do not appear to be within 1/16 inch in height relative to the adjacent pins, Gage (see 4.3.4) shall be used to determine the variation from the normal height. Reject any pin that varies by more than 0.063 inch max. from the normal pin height of the individual connector. The normal height of the individual connector pins shall be determined by calculating the average height from measurements taken of pins adjacent to the pin that appears to be defective. The forward metal edge of the connector shall be utilized as the reference surface of all measurements of pin heights.

**In the event socket inserts do not appear to be within 1/16 inch in depth relative to the adjacent socket, Gage (see 4.3.4) shall be used to determine the variation from the normal depth. Reject any insert that varies by more than 0.063 inch max. from the normal socket insert depth of the individual connector. The normal depth of the individual connector sockets shall be determined by calculating the average depth from measurements taken of sockets adjacent to the socket that appears to be defective. The forward edge of the connector insert shall be utilized as the reference surface for all measurements of socket depths.

4.3.2.1.6 Preparation for delivery (see 5).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
101.	Preservation, packing, and packaging not in accordance with MIL-STD-1169	Visual
102.	Marking not in accordance with MIL-STD-1169	Visual
Minor:		
201.	Evidence of poor workmanship (see 3.7)	Visual

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4.3.3 Testing. All tests shall be performed. Tests shall be performed in the order specified below:

Test	Test classification
a. Continuity	Minor
b. Pneumatic	Minor

None of these tests are considered destructive and samples so tested may be returned to the lot.

4.3.3.1 Continuity (see 3.4) and pneumatic (see 3.5). Each Cartridge Assembly and Instructor's Control Panel Assembly shall be tested in accordance with 4.4.1 and 4.4.2. If any item fails to comply with 3.4 and 3.5, the item shall be rejected and removed from the lot.

4.3.4 Acceptance inspection equipment.

4.3.4.1 Major characteristics and tests. Inspection and testing shall be performed with the acceptance inspection equipment, operating instructions, and calibration instructions specified on the Equipment Lists (EL) (see EL-8855810). When either of the conditions listed below exist, the contractor shall design required equipment or specify suitable commercial equipment and obtain approval prior to use on the contract.

a. Any deviation from equipment or procedures specified on the EL is desired (see 6.3).
b. The code "CD" is listed in the "Part or Identifying Number" column on the EL (see 6.1.2).

4.3.4.1.1 Additional Equipment. The following equipment shall also be required for these tests.

a. A power supply capable of supplying a current of two amps at 28 volts direct current (vdc).
b. A (T-229) T-4014 Tester.
c. A test adapter cable in accordance with Figure 1.
d. A regulated pressure nitrogen supply (350 pounds per square inch (psi)).

4.3.4.2 Minor characteristics and tests. Acceptance inspection equipment for minor characteristics is not specified on ELs. The acceptance inspection equipment, operating instructions and calibration instructions used by the contractor for minor classification inspection shall be approved by the Government representative authorized by the procuring agency.

4.4 Test methods. The tolerances specified herein are absolute with no allowance for test equipment inaccuracy. The tolerances used by the manufacturer shall be equal to the absolute tolerances less the accuracy tolerances of the test equipment used.

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4.4.1 Electrical tests (see 4.3.3.1).
4.4.1.1 T-4014 test. Complete test with malfunction.
4.4.1.1.1 Preparation for test.

NOTE: This test shall not be performed until 28 vdc power has been connected to P3 of the W5F Cable as follows: Plus connector to pins A, B, C, D, E, F, G, M. Minus connector to pins X, Y, Z, a, b, c, d, e.

4.4.1.1.1.1 Check receptacle J3 on cartridge to be certain that SAFE plug is properly inserted.

4.4.1.1.1.2 At this time, Mission Plug shall not be connected to receptacle J2 of the cartridge.

4.4.1.1.1.3 T-4014 AC PWR, Baro Set power S1, S4, S5 and S6 shall be in the OFF position.

4.4.1.1.1.4 Connect from T-4014 GND to earth ground with the W13-C ground strap.

4.4.1.1.1.5 T-4014 shall be connected to a 115 ± 10 volt, 60 Hz ac power source by means of the W3-H power cable, from receptacle J1 on the T-4014 to the power source.

4.4.1.1.1.6 Upon operation of AC power switch to the ON position, check that the AC power and defective lamps light and the test set motor starts.

4.4.1.1.1.7 When hinged access doors on the T-4014 have been opened, check that #3 lamp is lighted. The aforementioned lamps will light when pressed.

4.4.1.1.1.8 Operate AC PWR switch to OFF position. Observe the following: AC PWR #3 and DEFECTIVE lamps go out and test motor stops.

4.4.1.1.1.9 Connect W6H Test Cable from T-4014 (T-229) to the cartridge in the following manner.

Connect P1 to J3 on the T-4014.
Connect P2 to J2 on the T-4014.
Connect P3 to J6 on the Cartridge.
Connect P4 to J7 on the Cartridge.

4.4.1.2 Baroswitch setting procedures and limit test. Remote setting of Simulated Baroswitches is done using the T-4014 Test Set. The operation is performed using the same ground, power and test connections as were used previously on the electrical test. (Paragraph 4.4.1.1.1).

4.4.1.2.1 Upon operation of AC power switch to the ON position, check that the AC power and defective lamps light and the test set motor starts.

CAUTION: Do not operate Baro Set PWR Switch to the ON position unless switch S1 is the OFF position.

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4.4.1.2.1.1 Operate Baro Set PWR Switch to the ON position. (S2 is normally in Fuse Baro position.)

NOTE: If M-L Baro Indicator operates, allow it to come to rest before proceeding.

4.4.1.2.1.2 Operate switch, S-3 to the increase position until the indicator reads 25,000 ± 10 feet.

4.4.1.2.1.3 Operate switch, S-3 to the decrease position until the indicator reads 10,000 ± 10 feet.

4.4.1.2.1.4 Operate switch, S-3 to the decrease position until the indicator reads minus 3,000 ± 10 feet.

4.4.1.2.1.5 Operate switch, S-2 to the A.B. position and hold.

NOTE: If M-L Baro Indicator operates, allow it to come to rest before proceeding.

4.4.1.2.1.6 Operate switch S-3 to the decrease position until the indicator reads minus 3,000 ± 10 feet.

4.4.1.2.1.7 Operate switch, S-3 to the increase position until the indicator reads 10,000 ± 10 feet.

4.4.1.2.1.8 Operate switch, S-3 to the increase position until the indicator reads 25,000 ± 10 feet.

4.4.1.2.1.9 Release all switches.

NOTE: If M-L Baro Indicator operates, allow it to come to rest before proceeding.

4.4.1.2.2 Operate Baro Set power switch to the OFF position.

4.4.1.3 Continuity test.

NOTE 1: All installations for the above (paragraph 4.4.1.2) tests are required for this procedure.

NOTE 2: All switches on the Instructor's Panel to be in the "normal" position.

4.4.1.3.1 Warhead type.

4.4.1.3.1.1 Operate XS-XL switch on the Instructor's Panel to the XS position. The XS indicator on the T-4014 shall light.

4.4.1.3.1.2 Operate XS-XL switch on the Instructor's Panel to the XL position. The XL indicator on the T-4014 shall light.

4.4.1.3.2 Circuitry tests (T-4014).

4.4.1.3.2.1 Operate switches and observe that lights indicate as specified in Table IV.

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Table IV

J2 Insert	J3 Insert	PANEL SW-UP	S1 Position	READY LAMP	DEFECTIVE LAMP	PROCESSED LAMPS #1	#2	#3	#4	S.A. LAMP	S.S. LAMP
-	Safe	-	Off	Off	On	Off	Off	On	Off	Off	Off
-	Safe	-	1	On	Off	On	On	On	On	Off	Off
-	Safe	1-4	1	Off	On	On	On	On	Off	Off	Off
-	Safe	-	2	On	Off	On	On	On	On	Off	Off
-	Safe	2-1	2	Off	On	Off	On	On	On	Off	Off
-	Safe	2-2	2	Off	On	On	Off	On	On	Off	Off
-	Safe	2-3	2	Off	On	On	On	Off	On	Off	Off
-	Safe	2-4	2	Off	On	On	On	On	Off	Off	Off
-	Safe	-	3	On	Off	On	On	On	On	Off	Off
-	Safe	3-1	3	Off	On	Off	On	On	On	Off	Off
-	Safe	3-2	3	Off	On	On	Off	On	On	Off	Off
-	Safe	3-3	3	Off	On	On	On	Off	On	Off	Off
-	Safe	3-4	3	Off	On	On	On	On	Off	Off	Off
-	Safe	-	4	On	Off	On	On	On	On	Off	Off
-	Safe	4-1	4	Off	On	Off	On	On	On	Off	Off
-	Safe	4-2	4	Off	On	On	Off	On	On	Off	Off
-	Safe	4-3	4	Off	On	On	On	Off	On	Off	Off
-	Safe	-	5	On	Off	On	On	On	On	Off	Off
-	Safe	5-1	5	Off	On	Off	On	On	On	Off	Off
-	Safe	5-2	5	Off	On	On	Off	On	On	Off	Off
-	Safe	5-3	5	Off	On	On	On	Off	On	Off	Off
-	Safe	5-4	5	Off	On	On	On	On	Off	Off	Off
-	Safe	-	6	On	Off	On	On	On	On	Off	Off
-	Safe	6-1	6	Off	On	Off	On	On	On	Off	Off
-	Safe	6-2	6	Off	On	On	Off	On	On	Off	Off
-	Safe	6-4	6	Off	On	On	On	On	Off	Off	Off
-	Safe	-	7	On	Off	On	On	On	On	Off	Off
-	Safe	7-4	7	Off	On	On	On	On	Off	Off	Off
-	Arm	-	8	On	Off	On	On	On	On	Off	Off
-	Arm	8-1	8	Off	On	Off	On	On	On	Off	Off
-	Arm	8-2	8	Off	On	On	Off	On	On	Off	Off
-	Arm	8-3	8	Off	On	On	On	Off	On	Off	Off
S.A.	Safe	-	Off	Off	Off	Off	Off	Off	Off	On	Off
S.S.	Safe	0	Off	Off	Off	Off	Off	Off	Off	Off	On

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4.4.1.3.2.2 Operate the T-4014 AC PWR switch to OFF.

4.4.1.3.2.3 Connect regulated nitrogen supply to the cartridge support fitting adjacent to J5 of the cartridge.

4.4.1.3.2.4 Pressurize fitting to 10 psi max. and turn off tank valve.

4.4.1.3.2.5 Operate Instructor's Panel Static Probe system switch to the leak position. The pressure shall immediately fall off.

4.4.1.3.2.6 Insert Arm plug in J3 and S.S. plug in J2 of cartridge support respectively. The voltage meter shall show a maximum drop of 5 volts.

4.4.1.3.2.7 Insert Arm plug in J3 and S.A. plug in J2 of cartridge support respectively. The voltage meter shall show a maximum drop of 5 volts.

4.4.1.4 Circuitry tests (Launcher Operating Unit).

4.4.1.4.1 Continuity test.

4.4.1.4.1.1 Preparation for test.

4.4.1.4.1.2 Remove power from P3 of the W5-F Cable. Remove P3 and P4 of the T-4014 Cable from J6 and J7 on the Cartridge.

4.4.1.4.1.3 The continuity shall be as specified in Table V.

Table V
L. C. I. CIRCUITS

WH Switch Position	Panel Malfunction	J-2 Insert	J-3 Insert	W5-F - P3	Continuity W6-F - P2	Gnd
XS	-	-	-	-	C	Yes
XL	-	-	-	-	h	Yes
Either	-	S.S.	-	T	-	Yes
Either	-	S.A.	-	S	-	Yes
Either	-	-	-	-	B	Yes
Either	Ready	-	-	-	B	No
Either	-	-	Arm	R	-	Yes
Either	-	-	-	R	-	No
Either	-	-	-	-	D	Yes
Either	Safe	-	-	-	D	No
Either	-	-	-	-	E	No
Either	-	-	-	-	J	Yes
Either	-	-	-	-	e	No
Either	-	-	-	-	d	No
Either	-	-	-	-	K	No

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Table V (Con't)
L. C. I. Circuits

WH Switch Position	Panel Malfunction	J-2 Insert	J-3 Insert	Continuity W5-F - P3	W6-F - P2	Gnd
Either	-	-	Safe	A	-	Yes
Either	-	-	Safe	B	-	Yes
Either	-	-	Safe	C	-	Yes
Either	-	-	Safe	D	-	Yes
Either	-	-	Safe	E	-	Yes
Either	-	-	Safe	F	-	Yes
Either	-	-	Safe	G	-	Yes
Either	-	-	Safe	M	-	Yes
Either	-	-	Safe	X	l	
Either	-	-	Safe	Y	l	
Either	-	-	Safe	Z	l	
Either	-	-	Safe	a	l	
Either	-	-	Safe	b	m	
Either	-	-	Safe	c	m	
Either	-	-	Safe	d	m	
Either	-	-	Safe	e	m	

4.4.1.5 Fuse Baro Set.
4.4.1.5.1 Preparation for test.
4.4.1.5.1.1 Connect test adaptor cable as follows:
P1 to P2 of the W6F Cable.
P2 to P3 of the T-4014 Cable W6H.
P3 to P4 of the T-4014 Cable W6H.
4.4.1.5.2 Operate T-4014 power switch to the ON position.
4.4.1.5.3 Operate Baro Set power switch to the ON position.
NOTE: If M-1 Baro Indicator operates, allow it to come to rest before proceeding.
4.4.1.5.3.1 Operate switch, S-3 to the increase position until the indicator reads 1,000 + 10 feet.
4.4.1.5.3.2 Operate switch, S-3 to the decrease position until the indicator reads minus 3,000 + 10 feet.
4.4.1.5.3.3 Operate Baro set power switch to the OFF position.
4.4.1.5.3.4 Operate the T-4014 AC power switch to the OFF position.
4.4.1.5.4 Disconnect test adaptor cable from W6H and W6F Cables and store cable ends inside trainer.

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4.4.2 Pneumatic Tests.

NOTE: 1) Slight leakage of pop-off valves after closing shall not be cause for rejection.
2) Reduce tank pressure to 120 psi before accomplishing leakage test.

4.4.2.1 Operate System Control Valve and Bleed Valve on Instructor's Panel to OFF position.

4.4.2.2 Using regulated pressure nitrogen supply, charge panel fill until 300 psi gage reads 250-275 psi. Pop-off valve shall open.

4.4.2.3 Remove nitrogen supply from Panel fitting. Pop-off valve shall close at 200 psi minimum.

NOTE: Warhead pressure shall always be regulated before cartridge is pressurized.

4.4.2.4 Set system control valve to "Warhead" position.

4.4.2.4.1 Turn warhead regulator slowly in a clockwise direction, until 30 psi gage reads 25-30 psi. Pop-off valve shall open.

4.4.2.4.2 Turn warhead regulator in a counterclockwise direction until 30 psi gage reads 15 psi. Warhead pressure shall be 15 $\pm$ 2 psi.

4.4.2.4.3 Pop-off valve shall close at 18 psi min.

4.4.2.5 Set system control valve to cartridge position.

4.4.2.5.1 Turn cartridge regulator in a clockwise direction until 30 psi gage reads 5 psi. Cartridge pressure shall be 5 + 1 psi.

4.4.2.5.2 Turn warhead regulator and cartridge regulator in counterclockwise direction until they turn freely.

4.4.2.5.3 Set Bleed Valve to "Warhead" position; pressure shall drop to zero with control valve in warhead position.

4.4.2.5.4 Set control valve to "Cartridge" position; pressure shall drop to zero with bleed valve in cartridge position.

4.4.2.5 Record tank pressure.

4.4.2.6 After a minimum sixteen hour period, again check and record tank pressure.

4.4.2.6.1 Pressure loss shall not exceed 1 psi.

4.4.2.6.2 Set control valve and bleed valve to OFF position.

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5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A or C as specified (see 6.1).

5.1.1 Level A. The Cartridge Assembly and Instructor's Control Panel Assembly shall be preserved and packaged in accordance with instructions contained in MIL-STD-1169.

5.1.2 Level C. The Cartridge Assembly and Instructor's Control Panel Assembly shall be preserved and packaged in accordance with instructions contained in MIL-STD-1169.

5.2 Packing.

5.2.1 Levels A and C. The Cartridge Assembly and Instructor's Control Panel Assembly shall be packed in accordance with the applicable instructions contained in MIL-STD-1169.

5.3 Marking. Marking shall be in accordance with MIL-STD-1169.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

6.1.1 Procurement requirements.

a. Title, number, and date of this specification.

b. Selection of applicable level of packaging and packing required (see 5.1 and 5.2).

6.1.2 Contract data requirements (see 4.3.4.1). Data specified in 4.3.4.1 will be delivered as identified on a numbered DD Form 1664 when required on a DD Form 1423 incorporated in the contract.

6.2 Objective evidence. Records of contractor quality control and inspections which can be verified.

6.3 Acceptance inspection equipment (see 4.3.4.1). The contractor shall obtain approval from the Commander, Picatinny Arsenal, Dover, New Jersey 07801, ATTN: SARPA-QA-N-I.

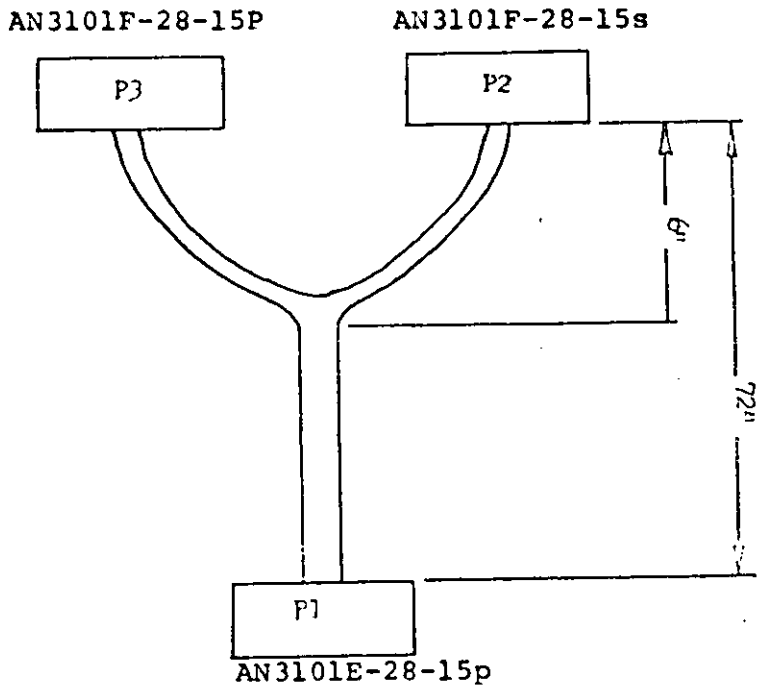
6.4 Supersession data. This specification supersedes the requirements of Purchase Description X-PA-PD-2129, dated 9 March 1962.

Custodian:
Army - PA

Preparing activity:
Army - PA

Project No. 1135-A198

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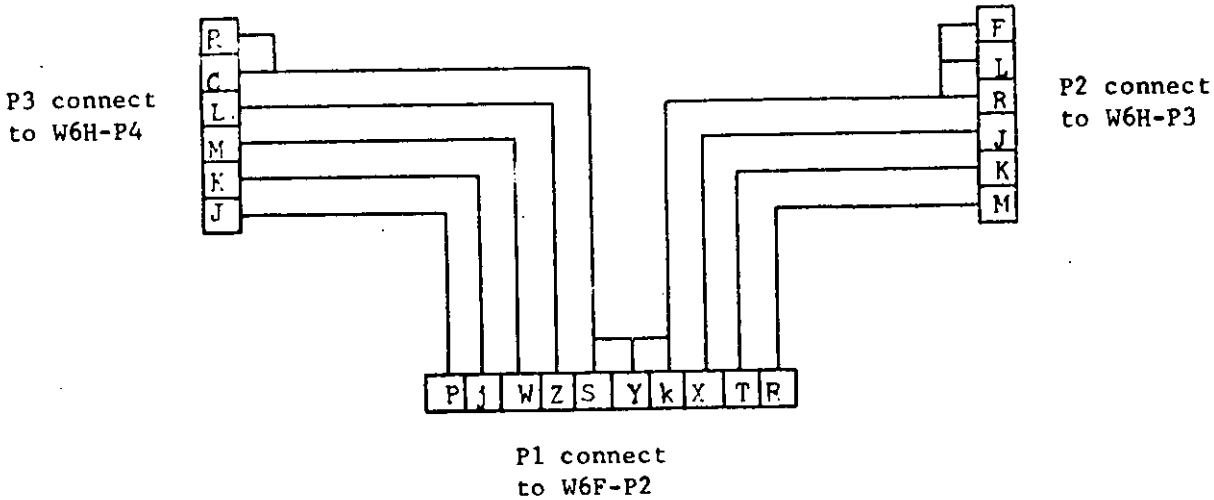


Figure 1

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FOLD

POSTAGE AND FEES PAID
DEPARTMENT OF THE ARMY
DOD – 314

U.S.MAIL

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

COMMANDER
Picatinny Arsenal
ATTN: SARPA-QA-N-Q
Dover, New Jersey 07801

FOLD

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

DD FORM 1 JAN 72 1426 REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

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